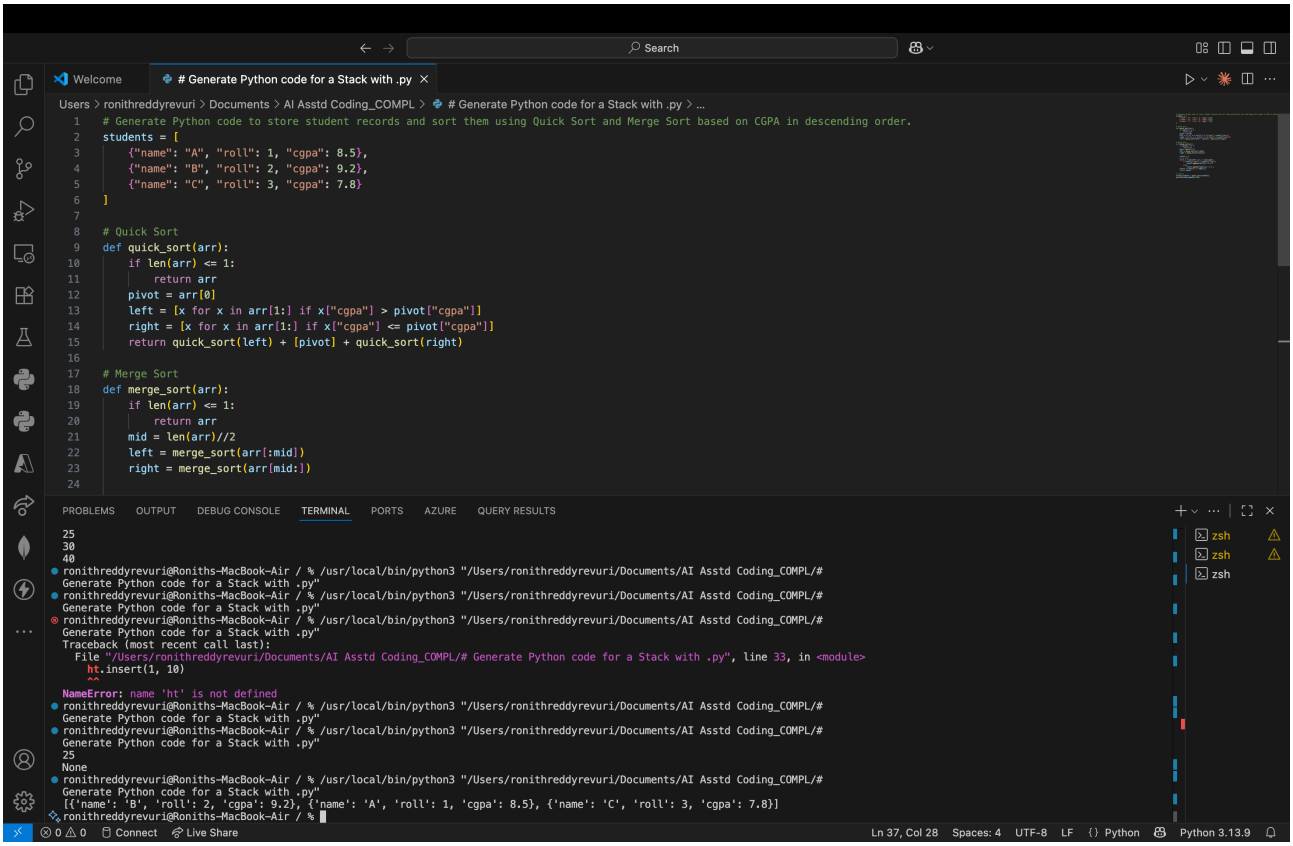
ASSIGNMENT-12.3

TASK 1 – SORTING STUDENT RECORDS

PROMPT :
GENERATE PYTHON CODE TO STORE STUDENT RECORDS AND SORT THEM USING QUICK SORT AND MERGE SORT BASED ON CGPA IN DESCENDING ORDER.

EXPLANATION. :
WHEN WE LOOK AT STUDENTS WE SORT THEM BASED ON THEIR CGPA. WE USE METHODS TO DO THIS SORTING OF STUDENTS BASED ON CGPA. FOR EXAMPLE QUICK SORT IS REALLY FAST WHEN WE ACTUALLY USE IT TO SORT STUDENTS BASED ON CGPA. ON THE HAND MERGE SORT IS VERY STEADY AND GIVES US STABLE RESULTS WHEN SORTING STUDENTS BASED ON CGPA.

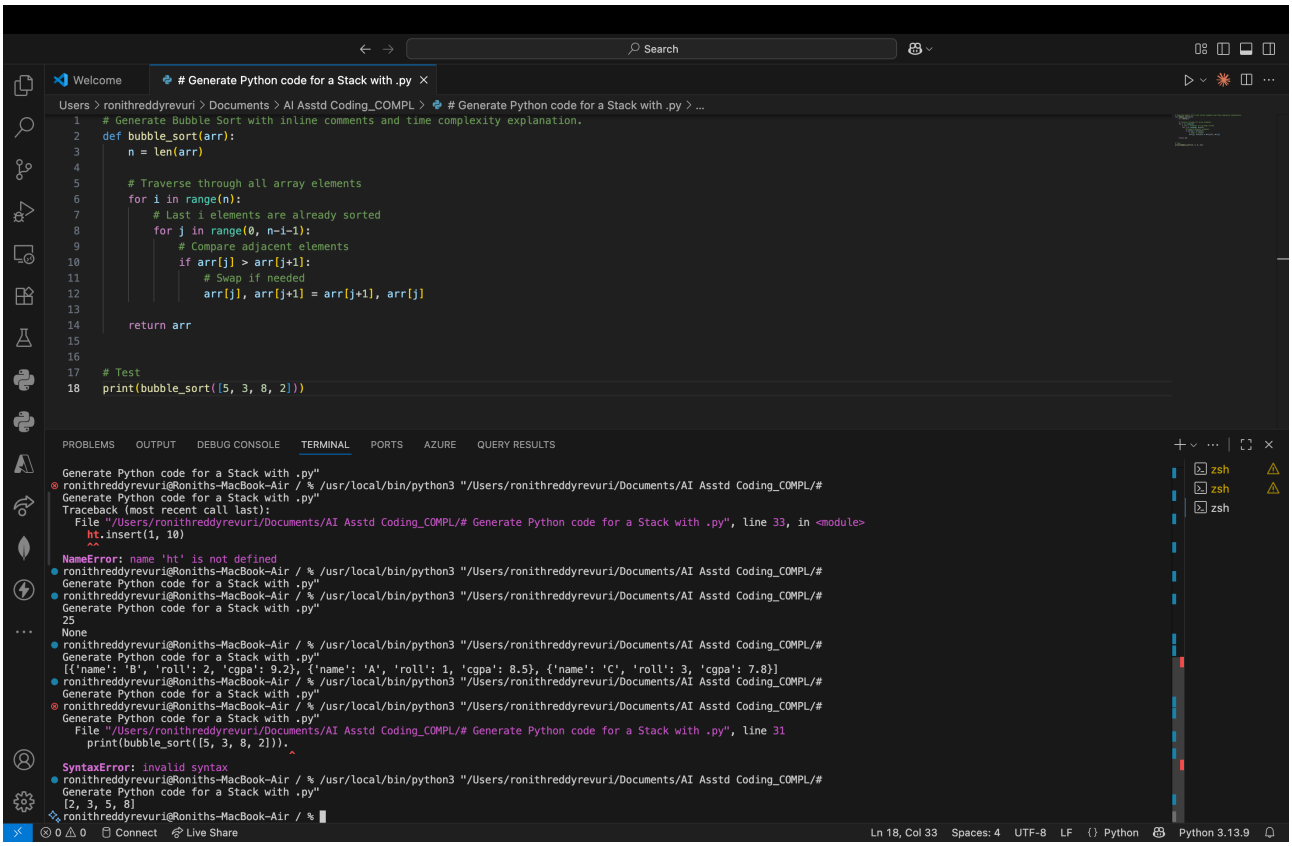


TASK 2 – BUBBLE SORT WITH AI COMMENTS

PROMPT :

GENERATE BUBBLE SORT WITH INLINE COMMENTS AND TIME COMPLEXITY EXPLANATION.

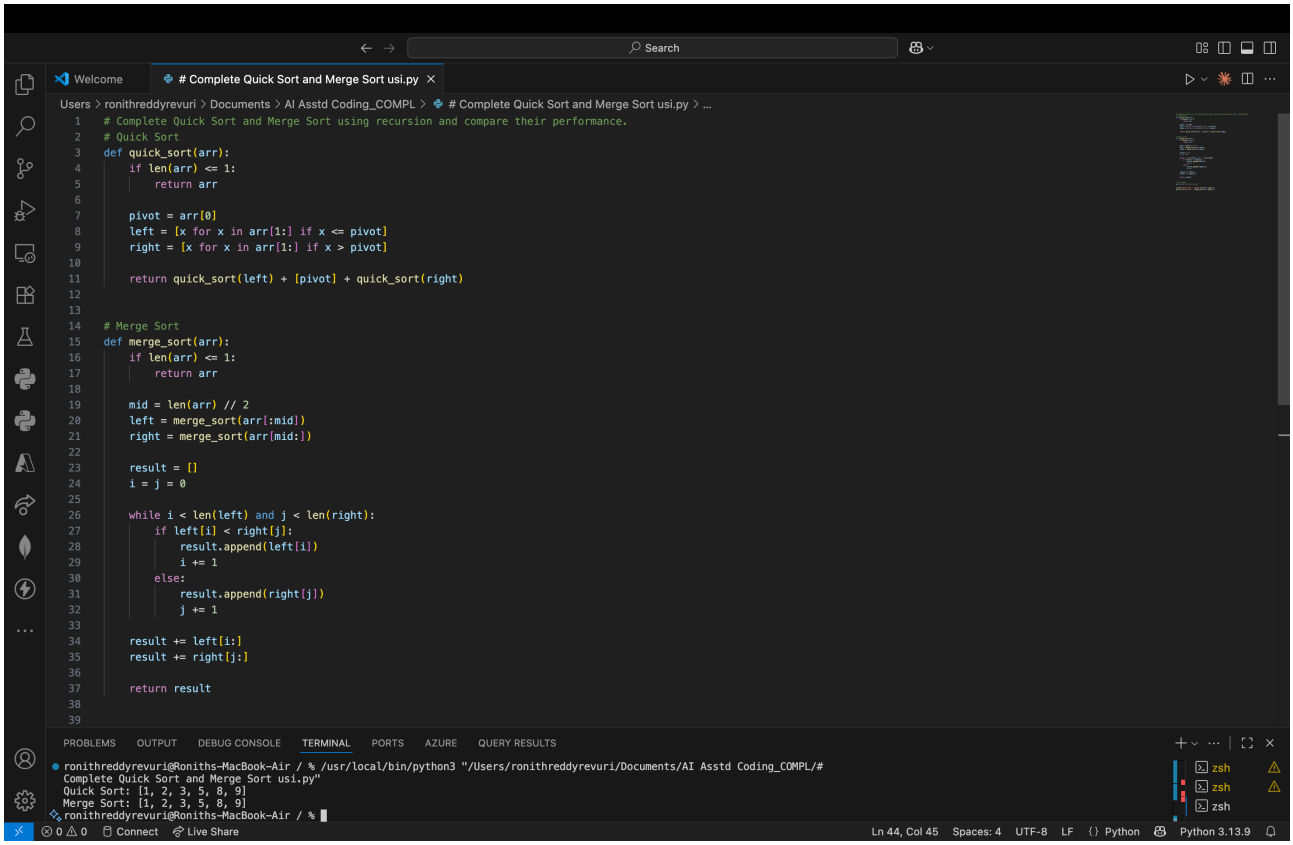
EXPLANATION :
BUBBLE SORT WORKS BY COMPARING AND SWAPPING ELEMENTS THAT'RE NEXT TO EACH OTHER.IT DOES THIS OVER AND OVER.THE TIME COMPLEXITY OF BUBBLE SORT IS O(N²). THIS MAKES BUBBLE SORT SLOW WHEN YOU HAVE A LOT OF DATA.IT IS NOT GOOD, FOR DATA SETS.BUBBLE SORT IS SIMPLE TO UNDERSTAND.IT IS USED FOR TEACHING.FOR REAL WORK OTHER SORTS ARE BETTER.



TASK 3 – QUICK VS MERGE SORT

PROMPT :
COMPLETE QUICK SORT AND MERGE SORT USING RECURSION AND COMPARE THEIR PERFORMANCE.

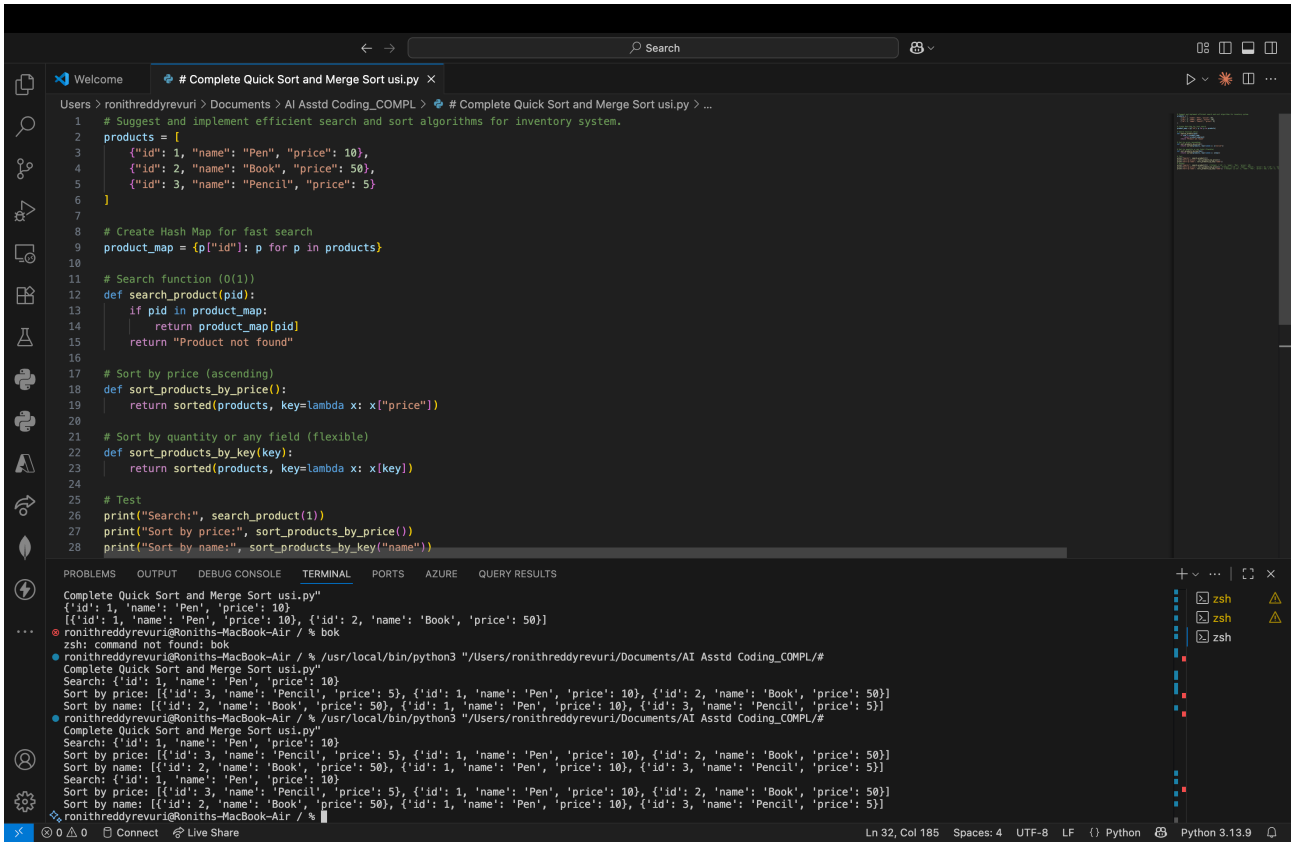
EXPLANATION :
BOTH OF THESE ALGORITHMS USE RECURSION. QUICK SORT USUALLY WORKS FASTER. MERGE SORT IS THE ONE THAT ALWAYS TAKES THE SAME AMOUNT OF TIME WHICH IS O(N LOG N) TIME.



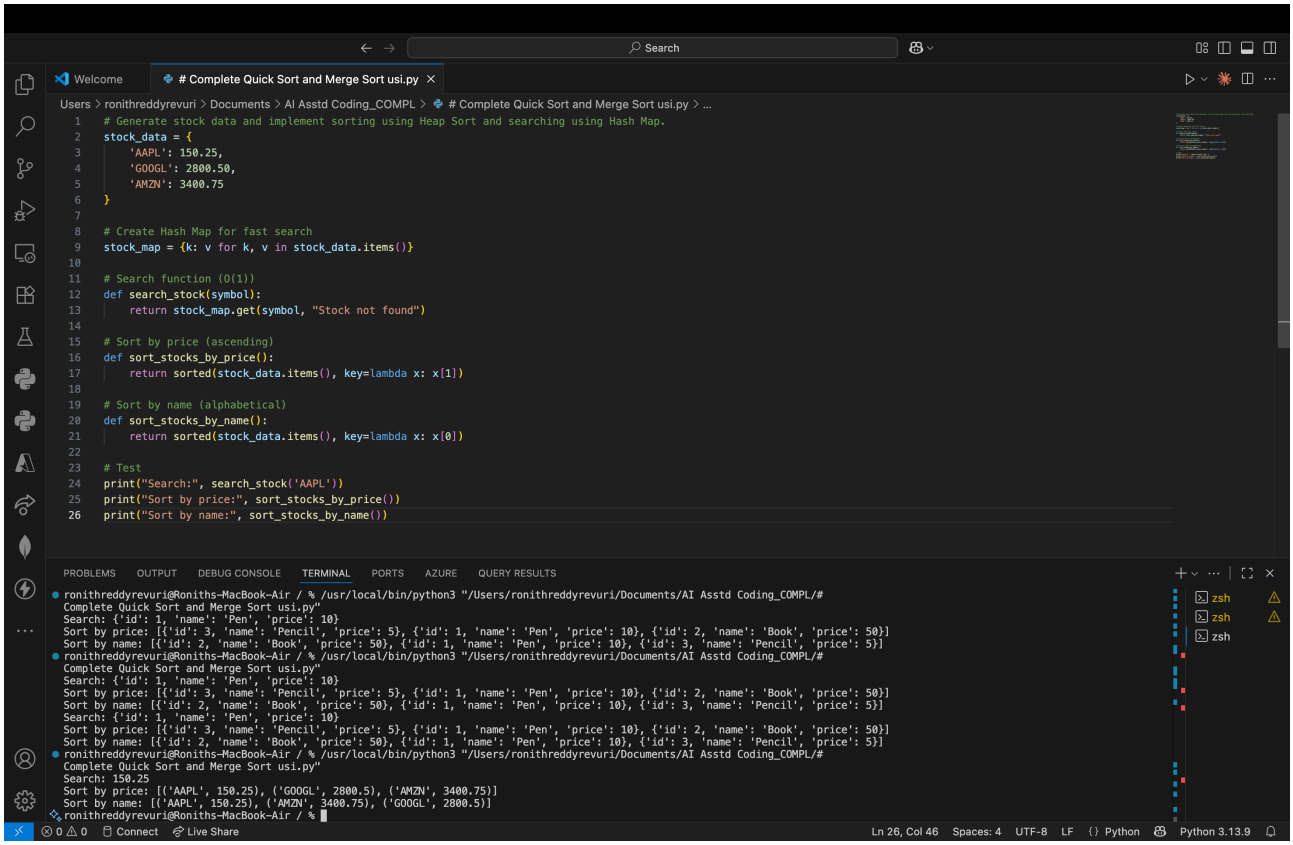
TASK 4 – INVENTORY MANAGEMENT SYSTEM

PROMPT :
SUGGEST AND IMPLEMENT EFFICIENT SEARCH AND SORT ALGORITHMS FOR INVENTORY SYSTEM.

EXPLANATION :
HASH MAPS ARE REALLY GOOD FOR SEARCHES. THEY CAN FIND THINGS IN ONE STEP, WHICH IS SUPER FAST. SORTING IS EASY TOO. WE USE BUILT-IN FUNCTIONS TO MAKE IT EFFICIENT. HASH MAPS ARE GREAT, FOR SEARCHING. THEY HELP US FIND WHAT WE NEED QUICKLY. SORTING USES THOSE BUILT-IN FUNCTIONS. IT MAKES THE WHOLE PROCESS FASTER.WE USE HASH MAPS FOR SEARCHING.THEY ARE REALLY FAST. SORTING IS DONE USING BUILT-IN FUNCTIONS. IT IS EFFICIENT THAT WAY.



TASK 5 – STOCK DATA SORTING & SEARCHING

PROMPT :
GENERATE STOCK DATA AND IMPLEMENT SORTING USING HEAP SORT AND SEARCHING USING HASH MAP.

EXPLANATION :
STOCKS ARE SORTED BY HOW THEY HAVE CHANGED IN PERCENTAGE.USING HASH MAPS HELPS TO FIND A STOCK BY ITS SYMBOL. THIS IS BECAUSE HASH MAPS ALLOW SEARCH.THE STOCKS ARE LISTED BASED ON THEIR PERCENTAGE CHANGE. HASH MAPS MAKE IT EASY TO LOOK UP A STOCK.